



Partial Membership Inference Attack

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Abstract

This dissertation investigates Membership Inference Attacks (MIA) across text, binary, and image datasets to assess their effectiveness when limited by partial data availability. Through experimental approaches including the Partial Noise Approach and Percentage Feature Approach, along with the innovative use of mutual information for feature selection, this study explores how Membership Inference Attack perform under constrained information scenarios. Although the exhaustive Enumerate Features Approach theoretically offers an optimal strategy, computational limitations restrict its practical evaluation.

Key findings reveal the potential of informed feature selection to enhance Membership Inference Attack methodologies and highlight the computational challenges of comprehensive enumeration. These insights contribute to the ongoing discussion on machine learning model privacy, underlining the importance of developing scalable, efficient privacy-preserving techniques. This work advances our understanding of the viability of Membership Inference Attack against incomplete data, setting a foundation for future privacy research in machine learning.

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Chapter 1

Introduction

In today's digital era, the widespread adoption of machine learning models in various sectors such as healthcare, finance, social networking, and security represents a significant advancement towards data-driven decision-making and automation. Nevertheless, this progress brings about its own set of vulnerabilities and obstacles, particularly in terms of data privacy. Among the cybersecurity risks, the Membership Inference Attack, as introduced by [9], stands out as a method specifically designed to compromise the privacy of the data utilized by these intelligent systems. The concept behind the Membership Inference Attack lies in the ability to ascertain whether a specific piece of data was part of the training dataset for a machine learning model. This attack not only jeopardizes the confidentiality of sensitive information but also sheds light on the often underestimated privacy vulnerabilities inherent in machine learning models. The repercussions of successful Membership Inference Attacks can range from the exposure of personal data in violation of privacy regulations to undermining the credibility and reliability of machine learning applications.

Membership Inference attack introduced by [1] first assumes the adversary would know the target model's structure and parameters and at least have a black box query access to the target model, which the adversary could input and investigate the output by the given input. Then the adversary would use output of the target model and train multiple shadow models that try to mimic target model's behaviour. After training the shadow models, the adversary observes their outputs for different inputs, focusing on the confidence levels or probabilities assigned to each class. These observations form the basis for training class-specific attack models. Each attack model aims to determine whether a data record was used in the shadow model's training set, leveraging the hypothesis that the shadow model's confidence in its predictions may differ for records it was trained on versus those it was not.

While the methodology behind Membership Inference Attack introduced by [9] demonstrates an approach to exploiting privacy vulnerabilities in machine learning models, it's important to note that the practical application of such attacks comes with significant challenges. Membership inference attack relies on several assumptions about the adversary's capabilities and the availability of data that

may not always be feasible in real-world scenarios. And, one of the assumptions underpinning Membership Inference Attacks is that the adversary has access to the complete data record for which they seek to determine membership in the target model’s training dataset. However, this assumption may not always align with real-world scenarios, where an adversary might only have partial information about a data record. In this dissertation, we explore the premise that Membership Inference Attacks can be effective even when the adversary has access only to partial or noisy data points. This investigation challenges the conventional belief that complete data is necessary for Membership Inference Attack, by examining the effectiveness of Membership Inference Attack under varied conditions of data completeness and noise, this work seeks to identify the boundaries within which such attacks remain viable.

Chapter 2

Background

2.1 Membership Inference Attack

Membership inference attacks introduced by [9] are a type of privacy attack against machine learning models that aim to determine whether a specific data record was part of the model's training dataset. These attacks pose a significant threat to the privacy of individuals whose data is used for training, as they can reveal sensitive information about their membership in a particular dataset.

To perform the attack, the adversary creates multiple "shadow models" that mimic the behavior of the target model. These shadow models are trained using synthetic data generated based on the target model's predictions. The adversary then trains an "attack model" using the shadow models' inputs and outputs, labeled as "members" or "non-members" of the shadow training sets.

The effectiveness of membership inference attacks is influenced by several factors, such as overfitting, high dimensional of the model's output, and skewed training data distribution. The authors propose potential defence mechanisms, including regularization techniques, coarsening and reducing the entropy of the model's outputs, and applying differential privacy.

Membership inference attacks highlight the importance of understanding and mitigating privacy risks associated with machine learning models, as they can reveal sensitive information about individuals through model predictions alone.

2.1.1 Shadow model

Shadow models are a key component of membership inference attacks. The purpose of shadow models is to imitate the behavior of the target model and allow the adversary to train the attack model with labeled input and output from the shadow model.

To train shadow models, it is assumed that the attacker would have access to the training data that is similarly distributed as target model's training data. There is few ways to gain access to shadow model training data, one approach is for the

attacker to have access to data from the same population as the target model’s training data, but not the exact data used to train the target model.

If the adversary does not have access to the training data and any knowledge of the training data distribution, then the adversary can obtain shadow model training data with the target model. This synthetic data generation process use hill climbing algorithm to search for the possible inputs by investigating the out from the target model find the input with the highest confidence score. By iteratively optimizing random inputs based on the target model’s predictions, the attacker can generate a synthetic dataset that mimics the distribution of the target model’s training data.

Once the shadow training data is generated, the attacker trains multiple shadow models on disjoint subsets of this data. For each shadow model, the attacker knows which records were used in training and which were not. The shadow models are then used to generate training data for the attack model: all pieces of data that is used to train the shadow models, the adversary will use them to query to shadow model and label the output with given input as "in" and "out". This labelled data is used to train the attack model to distinguish between predictions on training data versus other data.

The shadow training technique turns the complex relationship of if a piece of data is in the training dataset into a binary classification problem. This allows the attacker to effectively train the attack model to infer membership in the target model’s training dataset based solely on the target model’s predictions.

2.1.2 Attack model

The attack model is the final component of a membership inference attack. Its purpose is to infer whether a given input was part of the target model’s training dataset based on the target model’s prediction on that input.

To train the attack model, the attacker will use the labeled "in" and "out" labeled dataset generated by investigating shadow model’s output.

The methodology behind the attack model consists of assembling a suite of models, each tailored to one of the target model’s output classes. The rationale behind this approach is that the target model’s predictions exhibit unique distributions for each output class, influenced by the actual class of the input data. For every class, a binary classifier is developed, tasked with identifying whether a given prediction belongs to the 'in' category, indicating presence in the training data, or the 'out' category, signifying absences.

For executing the membership inference attack, the scenario is set where an attacker, with only black-box access to the target model, attempts to ascertain if a particular data record was utilized during the model’s training phase. The procedure involves submitting the data record to the target model and analyzing its prediction. This prediction is then input into the attack model’s relevant class-specific classifier, which renders a verdict of 'in' or 'out', indicating the likelihood of the data record’s prior inclusion in the training dataset.

2.1.3 Vulnerability

Membership inference attacks exploit the vulnerability of machine learning models by analyzing whether specific data was used in the model’s training set. This vulnerability is influenced by several key factors, including the distribution [3] and complexity of the training data [8], the number of classes in the dataset [10], the type of model used [10], the degree of overfitting [9], the difficulty of the task the model is trained to perform [8], and the dimensionality of the model’s output [8]. For instance, models trained on datasets with non-uniform class instances, a high number of classes, or complex data are more susceptible to these attacks [10]. Decision trees, due to their sensitivity to individual instances, and overfitted models, which tend to memorize training data, are particularly vulnerable [10]. Conversely, models like naive Bayes, which assume independence between features, show more resilience [10]. Furthermore, tasks that are inherently difficult or uncertain, leading models to memorize more training data, increase the risk of membership inference attacks [8]. This is especially true for models with high-dimensional outputs, where attackers can leverage the “ensemble effect” by aggregating information leakage across multiple output dimensions to improve attack accuracy [8]. Understanding these factors is crucial for developing defenses and privacy-preserving techniques in machine learning.

2.1.4 Defense

To defend against membership inference attacks, several strategies have been proposed. Regularization techniques, such as L2 regularization and dropout, aim to reduce overfitting, which is a major contributing factor to membership inference vulnerability [9] [7]. [6] introduced adversarial regularization, where the target model is trained to simultaneously minimize its prediction loss and an adversary’s inference gain. Differential privacy techniques, like DP-SGD [1], can provide provable privacy guarantees by adding noise during training, but often at the cost of model accuracy. [7] proposed model stacking, an ensemble method that trains multiple weak classifiers on disjoint data subsets to reduce overfitting. [5] introduced MemGuard, which perturbs the model’s prediction confidence scores to create adversarial examples that mislead the attacker’s inference model. Despite these efforts, preventing membership inference remains challenging, especially when the model is highly overfitted and the attacker has extensive knowledge. Balancing the tradeoff between model accuracy and membership privacy is crucial in developing effective defenses.

2.2 Partial Membership Inference Attack

In our study, we concentrate on exploring the limitations of membership inference attacks when the adversary possesses only incomplete information about the data point in question. Specifically, while a standard membership inference attack might assume an adversary attempting to discern whether a complete cat photo, consisting of 100 pixels, was part of the training dataset of the target model, our investigation seeks to understand the efficacy of such an attack under constrained conditions. We aim to ascertain whether the attack remains viable if the adversary has access to merely 80 pixels of the cat photo, thereby evaluating the attack’s success when operating with partial data.

Chapter 3

Related Work

Membership inference attacks have been studied extensively in recent years, with various approaches proposed to exploit vulnerabilities in machine learning models. However, most of these attacks operate under certain assumptions about the adversary’s knowledge and capabilities. Our research aims to explore the limitations of membership inference attacks when the adversary has incomplete information about the target data point. In this section, we discuss related work that also focuses on relaxing assumptions in MIA, albeit from different angles.

[3] propose two novel membership inference attacks, the Monte Carlo attack and the Reconstruction attack, which outperform previous approaches on both Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs). Their attacks demonstrate the vulnerability of generative models to membership inference, even under relaxed assumptions. The Monte Carlo attack only requires samples from the model and does not assume knowledge of the model type, while the Reconstruction attack is specifically designed for VAEs. In contrast, our research focuses on the scenario where the adversary has partial information about the target data point, rather than relaxing assumptions about the model itself.

[4] study membership inference under more realistic assumptions, such as skewed priors and threshold selection based on attack goals. They introduce a metrics based on positive predictive value (PPV) to capture privacy leakage in scenarios where the priors are skewed. Additionally, they propose a new attack strategy that evaluates the robustness of the target model’s predicted labels under input perturbations to infer membership. While their work considers realistic assumptions about the adversary’s goals and data distribution, our research specifically examines the impact of partial data knowledge on the success of membership inference attacks.

[2] introduce label-only membership inference attacks, which do not require access to the target model’s predicted confidence scores. Their attacks exploit the robustness of the model’s predicted labels under input perturbations, such as data augmentations and adversarial examples, to infer membership. They demonstrate that these label-only attacks can match the performance of at-

tacks that rely on access to confidence scores. Although their work does not directly address the issue of partial data knowledge, it highlights the effectiveness of attacks under restricted access to model outputs, which is relevant to our research.

[7] propose membership inference attacks that gradually relax assumptions about the adversary’s shadow model design and training data distribution. They introduce a data transferring attack that allows the adversary to use a different dataset to train the shadow model, and they also propose an attack that does not require training any shadow model. While their work focuses on relaxing assumptions related to the adversary’s model training capabilities and data distribution knowledge, our research specifically investigates the impact of partial data knowledge on the success of Membership Inference Attack.

In summary, while the referenced papers explore relaxing assumptions in membership inference attacks from various perspectives, such as model type, adversary goals, data distribution, and access to model outputs, our research specifically focuses on the limitations of these attacks when the adversary has incomplete information about the target data point. By investigating the effectiveness of membership inference attacks under partial data knowledge, we aim to provide a complementary perspective on the robustness of these attacks and the privacy risks associated with machine learning models.

Chapter 4

Methodology

4.1 Machine Learning Models

We have experimented Partial Membership Inference Attacks across different machine learning models, including Multinomial Logistic Regression and Multi-Layer Perception (MLP) Neural Networks.

4.1.1 Multinomial Logistic Regression

Multinomial logistic regression extends the traditional logistic regression model to handle multi-class classification tasks involving more than two categorical outcomes. This modification broadens the scope of logistic regression beyond binary classification, enabling it to address scenarios where the response variable can assume multiple discrete values. At the heart of multinomial logistic regression lies the softmax function, which is an extension of the logistic or sigmoid function. The softmax function transforms a vector of real-valued inputs into a probability distribution across multiple classes, ensuring that the sum of probabilities for all classes equals 1. This property makes it well-suited for predicting the likelihood of each potential class.

Within this framework, the model utilizes the softmax function on a linear combination of input features to predict the probability distribution across all classes. The model's coefficients, which represent the impact of each input feature on the probability of each class, are optimized using techniques such as gradient descent. This optimization process aims to minimize the disparity between the model's predicted probability distribution and the actual class outcomes, enhancing the model's ability to capture the intricacies of the data.

Multinomial logistic regression is valued for its simplicity, interpretability, and flexibility in multi-class scenarios. It offers a probabilistic perspective on classification, allowing for an examination of how different features influence the likelihood of class membership. However, it assumes a linear relationship between the log odds of classes and input features, an assumption that may not hold universally across all datasets.

We selected multinomial logistic regression as the initial model for evaluating

partial-membership inference attacks due to its simplicity and transparency. This choice enabled us to establish a more fundamental and clearer understanding.

4.1.2 Multilayer Perception Neural Network

A Multilayer Perceptron (MLP) is a feedforward artificial neural network that consists of multiple layers of nodes, including an input layer, hidden layers, and an output layer. Each node in a layer is connected to every node in the subsequent layer, creating a dense network of weighted connections. MLPs utilize supervised learning with backpropagation to adjust weights based on the variance between actual and predicted outputs.

The significance of MLP lies in their hidden layers and the incorporation of non-linear activation functions like Rectified Linear Unit (ReLU), sigmoid, and hyperbolic tangent, enabling the network to capture complex, non-linear relationships within data. Through the training process, which involves gradient descent or its variations, the network iteratively optimizes weights to minimize a loss function that quantifies prediction error.

The complexity of MLPs, determined by the number of hidden layers and neurons, is linked to the network’s capacity to represent intricate data patterns, while also impacting computational requirements and the risk of overfitting. Overfitting, a phenomenon where a model closely fits the training data but performs poorly on new data, poses a significant challenge in MLP design.

We have opted for the Multilayer Perceptron (MLP) as our model for conducting partial membership inference attack experiments due to its complexity and susceptibility to overfitting if not carefully calibrated. MLP with their deep architectures involving multiple hidden layers, possess the ability to capture highly complex data patterns. However, this complexity also makes them prone to overfitting, especially when the model memorizes the training set along with the underlying data patterns. This susceptibility to overfitting is crucial in the context of membership inference attacks, as models that memorize training data are more vulnerable to such attacks.

In the realm of privacy and security, an overfitted model serves as a more realistic situation for adversaries, mirroring potential real-world situations where models might not be optimally regularized or tuned. Overfitted models will "memorize" the training data characteristics deeply, and provide a fracture for membership inference attacks, making them ideal candidates for assessing the robustness of data privacy techniques. Through the application of MIA to an MLP that is prone to overfitting, we aim to simulate an environment where the model’s vulnerability to such attacks is heightened, thus providing insight into how data privacy might be compromised in practical scenarios.

4.2 Dataset

In our research, we have meticulously sought to explore the extent and limitations of Partial Membership Inference Attacks. This exploration has been conducted across a diverse array of dataset types, encompassing image datasets,

text datasets, and binary datasets. Through this approach, we aim to uncover nuanced insights into how these attacks vary in effectiveness depending on the nature of the data.

4.2.1 Binary Dataset: Purchase

For our experiments on binary data, we utilized the Purchase dataset which is used by [1], which is a simplified version derived from Kaggle’s ”Acquire Valued Shoppers Challenge.” This dataset captures the shopping history of individuals, featuring 197,324 data points. Each data point consists of 600 binary features, where each feature indicates whether the individual has purchased a specific product. However, Purchase Dataset does not contain any prediction classes, by following [1] we used a clustering algorithm named k-means cluster the data points into 2, 10, 20, 50, 100 classes, which each class is used as prediction classes for the Purchase dataset.

K-means

K-means is a clustering algorithm that groups data into K distinct clusters by minimizing within-cluster variances. It starts by selecting K initial centroids, then assigns data points to the nearest centroid, forming clusters. Centroids are recalculated as the mean of all points in each cluster, and this process repeats until a stopping criterion is met. Its efficiency is influenced by the initial centroid positions and the choice of K, assuming spherical clusters of similar size. For the Purchase dataset, K-means clustered data points with 600 binary features into classes (2, 10, 20, 50, 100), serving as prediction classes.

4.2.2 Image Dataset: CIFAR 10

For our experiments on image data, we utilized the CIFAR-10 which is also used by [1]. The CIFAR-10 dataset are pivotal in machine learning and computer vision research, serving as benchmarks for evaluating image classification algorithms. Both datasets contain 60,000 images each, with dimensions of 32x32 pixels in full color. CIFAR-10 is organized into 10 classes, each containing 6,000 images, representing everyday objects such as airplanes, cars, birds, and dogs. This division allows for a broad spectrum of basic object recognition tasks.

4.2.3 Text Dataset: Newsgroups 20

For our experiments on text data, we utilized the Newsgroups 20 dataset from scikit-learn. The 20 Newsgroups dataset is a collection widely used in machine learning and natural language processing for text classification, clustering, and information retrieval experiments. this dataset comprises around 20,000 newsgroup documents, partitioned evenly across 20 different newsgroups. Each newsgroup corresponds to a distinct topic, covering a broad range of subjects from sports and science to politics and religion. This diversity in content makes the dataset an excellent resource for tasks requiring the understanding and categorization of text based on its thematic material. We also reprocess the news

dataset by removing headers, footers and quotes from the document as `skcitlearn` recommended for machine learning then factorize it using `Skcitlearn's tfdvectordize`.

4.3 Membership Inference Attack

In our research, we closely followed the methodology established by [1] as base for Partial-Membership Inference Attacks by involving training several shadow models on data subsets similar to the original training data of the target model. These shadow models mimic the target model's behavior, allowing us to collect data on how the model responds to both seen and unseen inputs. This data then trains attack models, for which there is a model for each prediction class.

4.4 Partial Membership Inference Attack

In our investigation on Partial Membership Inference Attack, we utilized a traditional MIA framework to assess the effectiveness of the attack when faced with limited data availability. The objective of this study was to ascertain whether the attack could accurately determine if a specific data point was part of the training dataset of the target model, even with only part of the features of that piece of data is at hand. We developed four distinct strategies to replicate different real-world scenarios where only fragments of data may be accessible, with some strategies aiming to enhance the attack's precision in specific circumstances. These strategies were tested across a range of machine learning models and data formats, ensuring consistency in approach while tailoring the implementation to the unique attributes of each data type.

4.4.1 Approach 1: Partial Noise

In our "Partial Noise" methodology, the emphasis lies on the deliberate application of noise to particular data points, with the objective of determining their inclusion within the training dataset of a machine learning model. This calculated intervention replicates real-life situations where the reliability or authenticity of individual data points could be jeopardized, enabling an evaluation of the impact of such disturbances on the efficacy of Membership Inference Attacks.

The methodology involves different strategies tailored to different data types: binary, image, and text data. For binary data, the approach is to flipping the bits of a designated percentage of features within the data point, thereby introducing binary-level noise. This simulates conditions where binary features may be subject to errors or disturbances, affecting the data point's original representation.

In the case of image data, we employ Gaussian noise to randomly subset of features within the targeted data point. This method mimics common issues in image data collection and processing, such as sensor noise or transmission errors, affecting a portion of the image's pixels and potentially altering its characteristics.

For text data, which undergoes vectorization to convert textual information into a numerical format suitable for machine learning models, Gaussian noise is similarly applied to a portion of the features of the data points. This reflects the variability and potential inaccuracies in the numerical representation of text, which could arise from errors in data processing or the inherent ambiguity in linguistic expressions.

The objective of applying noise to the data point is to analyze the limitations within which Membership Inference Attack can effectively determine the membership status, even when the data points are partially modified. This approach plays a crucial role in comprehending the robustness of MIA against flawed data, providing valuable insights into protecting sensitive data from unauthorized inference and improving the overall privacy measures in machine learning applications.

4.4.2 Approach 2: Percentage Features

In our "Percentage Feature" approach, the selection of features from each datapoint is performed randomly, aiming to evaluate the efficacy of Membership Inference Attacks under conditions of partial data availability. This randomized feature selection method is designed to assess the resilience of MIA when there is only a fraction of the feature of the data point.

By using a randomized selection method, we ensure that our feature combinations include a diverse range of possibilities. This allows us to consider situations where an attacker may not have systematic or predictable access to certain features due to privacy measures, data issues, or limitations in data collection. Introducing randomness reflects the uncertainties faced in real-world scenarios when acquiring and processing data. As a result, our evaluation achieves a higher level of resilience and thoroughness.

The main idea behind this strategy is to change the ratio of characteristics used from each data point. This allows us to systematically examine and identify the point at which Membership Inference Attacks (MIAs) can accurately determine if a data point was part of the training dataset for a machine learning model. This approach provides a quantifiable measure of how the inclusiveness of feature sets impacts the success of MIAs, thus highlighting the important balance between data availability and vulnerability to privacy breaches.

Applying this randomized feature selection strategy across different data types—binary, image, and text—enables us to adapt the investigation to the unique attributes and challenges posed by each type of data. This exploration is invaluable for understanding the minimal informational requirements for conducting successful Membership Inference Attack.

4.4.3 Approach 3: Feature Selection

In our "Feature Selection" approach, we use mutual information to select the most informative features from the data point, aiming to find out its membership in the training dataset more accurately. This method is specifically designed to improve the effectiveness of Membership Inference Attacks beyond with our "Percentage Feature" approach. By using the same percentage of features in

both methods, our objective is to demonstrate that a targeted selection based on mutual information can produce superior results in validating the attack.

This strategy hinges on the premise that not all features contribute equally to the model’s predictions. Mutual information allows us to identify and focus on those features that share a significant amount of information with the outcome, suggesting they play a pivotal role in the model’s decision-making process. The selection of features with the highest mutual information scores is expected to provide more value for the MIA, as it concentrates the attack on the data aspects most influential to the model’s behaviour.

This approach contrasts with the "Percentage Feature" method, where features are selected randomly, regardless of their contribution to model predictions. By comparing the effectiveness of these two strategies using an identical percentage of features, we aim to illustrate the added value of informed feature selection. The hypothesis is that by concentrating on the most informative features, we can achieve a more precise and efficient inference attack.

4.4.4 Approach 4: Enumerate Features

In our "Enumerate Features" approach, we adopt a strategy where a specified percentage of features is selected and treated as if they were missing, to simulate scenarios where complete information is not available (percentage features). This involves enumerating a vast of combinations regarding which features are considered missing, thereby allowing us to rigorously test the resilience of Membership Inference Attacks against incomplete data. The premise of this approach is to mimic various real-world situations where data might be partially available due to reasons like privacy settings, data collection limitations, or loss of data and to improve the attack.

The procedure begins with the random selection of a subset of features, designated as "missing," and proceeds to enumerate a spectrum of possibilities for the missing features. This extensive exploration enables us to evaluate the feasibility of inferring membership even when data is partially obscured or unavailable. Essentially, by simulating a diverse range of scenarios with different combinations of missing features, we aim to create conditions under which an attacker might still manage to reconstruct or infer the missing information, thereby executing an attack as if they had access to the full data point. This strategy allows for the investigation of whether certain configurations of missing features can still provide sufficient information for a successful MIA, potentially revealing "guessable" values for the missing features that closely mimic the original dataset.

However, it’s important to acknowledge the computational complexity inherent in this approach. Enumerating a vast array of feature combinations and assessing each for MIA susceptibility requires substantial computational effort and resources.

Chapter 5

Implementation

In this chapter, we delve into the detailed experimental framework of Partial Membership Inference Attack and its associated approaches. note(I implemented the base framework of Membership Inference Attack as developed by my supervisor Pascal Berrang for this research)

5.1 Base Frame work: Membership Inference Attack

For a Partial Membership Inference Attack, the core methodology mirrors that of a standard Membership Inference Attack, where the objective is to deduce whether specific data points were part of the model’s training set. The distinction, however, comes into play during the validation phase. In a Partial Membership Inference Attack, the validation process involves modifying the data points in question before assessing their membership status. This modification typically consists of omitting or altering a subset of features within the data points, simulating scenarios where complete information might not be available or where features have been intentionally obscured. In our experiment, we have opted to utilize the methodology of the traditional Membership Inference Attack as outlined by [1], and here we will describe the implementation.

5.1.1 Labeled dataset

In preparation for the framework for membership inference attack, we use a fundamental data structure in code to represent a labeled dataset. It is defined as a dataclass and consists of two main attributes: X and y. The X attribute represents the feature vectors or input data, while the y attribute represents the corresponding labels or target values. In code this data structure is Labeled dataset.

The LabeledDataset class in the described setup is designed to efficiently manage datasets with both features (X) and labels (y). Upon initialization, it takes two arguments: X for the input samples and y for their corresponding labels. A post-initialization step to ensures the integrity of the dataset by asserting that the

lengths of X and y are equal, thus verifying that each sample has an associated label. For convenience, this class can be constructed directly from a Pandas DataFrame using the `from_pandas` static method, which requires the dataframe and the name of the column that contains the labels. This method separates the feature vectors and labels, facilitating the creation of a `LabeledDataset` instance directly from tabular data.

Label management is an essential feature, where the `transform_label` method allows for the conversion of labels to a numeric format, utilizing scikit-learn’s `LabelEncoder`. This conversion is crucial for machine learning tasks that require numerical labels. The method not only transforms the labels but also returns the fitted encoder, enabling the reverse transformation if needed.

The class provides functionalities for dataset management, including methods to determine the dataset’s length using `len` and to split the dataset. Splitting can be done either proportionally into training and testing sets via the `split` method, which relies on scikit-learn’s `train_test_split` function for a stratified split, or at a specific index using the `split_at` method, yielding two subsets of the dataset. Additionally, the `sample` method offers the capability to generate a new dataset of a specified size through sampling, which is useful for creating balanced or smaller datasets for experimentation or cross-validation.

Lastly, the `num_classes` method returns the count of unique classes or labels present in the dataset, providing insight into the dataset’s diversity and aiding in tasks like classification where the number of output neurons or the configuration of a machine learning model may depend on the number of unique labels.

5.1.2 Configuration

In preparing for our partial Membership Inference Attack experiment, we implement a crucial configuration step that plays a crucial role in configuring the dataset splitting for the Membership Inference Attack evaluation. It ensures that the specified sizes for training and testing sets are compatible with the available dataset and the desired proportions. By validating the configuration, it helps prevent invalid or inconsistent settings that could lead to errors or unexpected behaviour during the Membership Inference Attack evaluation process. The class we used for configuration is `MIAconfig`. There are four attributes in configuration

- **Absolute target size:** An integer representing the absolute number of samples allocated to the target models.
- **Absolute training size:** An integer representing the absolute number of training samples used to train a single classifier.
- **Absolute testing Size::** An integer representing the absolute number of testing samples used to test a single classifier.
- **Training proportion:** A float represents the proportion of samples allocated for training.

First, we use the `post_init` method to assert the attribute training proportion is between 0 and 1, to ensure the training proportion represents a valid proportion within the expected range. Then we use `validate` method to validate the

configuration based on the total number of samples in the dataset. It performs several assertions to ensure that the specified sizes for training and testing sets are compatible with the dataset size and the chosen proportions. If any of the assertions fail, it indicates that the dataset and proportions do not allow for the desired training and testing set sizes. Finally, we use from proportion method, It takes the total number of samples, the relative target size (between 0 and 1), the absolute training size, the absolute testing size, and the training proportion as arguments. It calculates the absolute target size based on the relative target size and validates the training and testing sizes against the dataset size and proportions. If the sizes are valid, it returns a new MIAConfig instance with the calculated absolute target size and the provided training and testing sizes.

5.1.3 Environment

For the environment which is the main component for the membership inference attack, it is used to set up and manages the evaluation for membership inference attack. For environment it takes a labeled dataset, a target model, shadow model, attack models and an configuration instance as input and provides methods for training test and performing membership inference attack. Which the class we use for creating the environment is MIAEnvironment.

initialization

The initialization process has few steps:

1. The dataset's labels are transformed using a label encoder to ensure they are numeric and ascending.
2. The target model, shadow models, and attack models are cloned to create separate instances for the evaluation.
3. The Configuration we mention in previous section is validated against the dataset size to ensure compatibility.
4. The dataset is split into target and shadow parts based on the absolute target size specified in the configuration.
5. The target and shadow datasets are further split into training and testing sets according to the training proportion in the configuration.
6. The training and testing sets for each model are resampled based on the absolute training size and the absolute testing size from the configuration.

After the initialization the environment will start the process of membership inference attack.

Training and Testing:

The environment provide several methods for training and test the target, shadow and attack models, these methods are used to facilitate the evaluation of membership inference attack.

- Train basic model: This method is used to train target and shadow models, with their respective dataset
- Test basic model: This method is used to test the target model and shadow models using their respective testing datasets.
- Train Attack model: This method is responsible for training the attack models using the shadow models' predictions on their training and testing datasets. It will first asserts that target models and shadow models have been trained or not then if there is no attack model training data or not if there is none it will call a utility function to calculate the training data set for attack model. Finally we used the calculated dataset to train the attack model.
- Test attack model: This method is used to test the attack models using the target model's predictions on its training and testing datasets.

These training and testing methods provide a structured way to train and evaluate the target, shadow, and attack models in the membership inference attack scenario. The train basic models and test basic models methods handle the training and testing of the target and shadow models, while the train attack models and test attack models methods focus on training and testing the attack models using the shadow models' predictions.

Attack dataset calculations

After shadow models are trained and when we are training attack model with no attack dataset yet, we will use shadow models to calculate the attack dataset to train attack models. The attack dataset calculations is a utility function for train attack models to create the attack dataset for each class label. In code this utility function is get attack dataset.

1. It first starts by initializing variables to store attack datasets. For each class present in the original dataset, it creates separate lists to hold the attack inputs (attack X) and their corresponding labels (attack y), indicating membership or non-membership.
2. Then it iterates over the provided shadow models and their corresponding datasets. For each model-dataset pair: It samples a subset of data points, ensuring the size matches data size, which represents the desired number of data points for each class in the attack dataset and uses the shadow model to predict class probabilities (posteriors) for both the training and testing subsets of the sampled dataset.
3. Then For each data point in the sampled datasets it identifies the data point's true class and adds the model's predicted posteriors to the attackX list corresponding to that class and labels data points from the training subset as "IN" (indicating they are members of the training set) and those from the testing subset as "OUT" (indicating they are not members of the training set). These labels are added to the attacky list for the respective class.
4. After processing all models and their datasets, the method transforms the collected attack X and attack y lists into labeled datasets. It creates a

LabeledDataset instance for each class, comprising the attack inputs and their membership labels.

5. Finally, the method returns a list of LabeledDataset instances, one for each class. Each dataset can then be used to train or evaluate an attack model that predicts membership for that specific class.

This method plays a crucial role in the MIA process by ensuring that attack models are trained and evaluated on data that accurately reflects the task of distinguishing between members and non-members of a model’s training set. By constructing these datasets based on predictions from shadow models, it simulates the scenario where an attacker has access to a model similar to the target model and uses it to infer membership information.

Validate membership

After attack models are trained we will perform membership inference on one or more input samples. We will first take a set of input features and determine whether each samples is likely to be a member of the target model’s training dataset. Which it is **is member** method in the code.

1. The method takes an input X, which can be a single data point or multiple data points (array-like structure). The method first uses the target model to predict the class probabilities (posteriors) for X.
2. Then it determines the predicted class for each data point by finding the index of the maximum probability in the predicted class probabilities. This step identifies which attack model (corresponding to a class) to use for each data point in X.
3. After that it groups the predicted posteriors by their predicted class indices to efficiently query the corresponding attack models. This involves sorting the posteriors and indices by class indices, finding the unique class indices, and splitting the posteriors accordingly.
4. Then, For each group of posteriors (each corresponding to a class), the method uses the corresponding attack model to predict whether each data point in the group was part of the training set (member) or not (non-member). The attack model outputs probabilities for membership, from which a boolean decision (member or non-member) and a posterior probability of membership are derived. Finally, it aggregates the boolean decisions and the membership posterior probabilities for all data points across all classes. It returns these as 1D vectors, where each element corresponds to a data point in the input X.

5.2 Approach 1: Partial Noise

In the ”Partial Noise” approach, our objective is to introduce noise into specific pieces of data that we aim to validate for their membership within the training dataset. The application of this approach varies slightly across different data types, reflecting the inherent characteristics and requirements of each type of data.

Binary data

In the implementation of noise addition for binary data, we utilized two parameters: the percentage of features across which to apply noise and the probability of flipping bits for the selected features. The procedure involves randomly selecting a subset of features, based on the specified percentage, and then applying bit flipping to each bit within this subset with the defined probability. Specifically, a bit with a value of 0 is flipped to 1, and a bit with a value of 1 is flipped to 0, according to the stipulated probability. This method is systematically applied to introduce noise into the binary data, with the extent of noise controlled by the adjustable parameters of feature percentage and bit-flipping probability.

Image data

In the implementation of noise addition for image data, we also utilized two parameters: the percentage of features across which to apply noise and the noise level of Gaussian Noise for adding noise to the selected features. The procedure involves randomly selecting a subset of features, based on the specified percentage, and then applying Gaussian Noise to the selected columns of features. This method is systematically applied to introduce noise into the image data, with the extent of noise controlled by the adjustable parameters of feature percentage and Gaussian Noise level.

Text data

In the implementation of noise addition for text data, since the text data is vectorized we also utilized Gaussian noise, with two parameters like the implementation of image data: the percentage of features across which to apply noise and the noise level of Gaussian Noise for adding noise to the selected features. The procedure is the same as image data by randomly selecting a specified percentage of features and then applying the stated noise level of Gaussian Noise to the selected columns of features.

5.3 Approach 2: Percentage Features

In the "Percentage Features" approach, our primary goal is to assess the impact of utilizing only a subset of features from each data point for validating its membership in the training dataset. This approach is uniformly applied across all data types, be it binary, image, or text data, to maintain consistency in the experimental design and ensure the comparability of results. The implementation process begins by randomly selecting a predetermined percentage of features from each data point. These selected features are then treated as if they are missing, effectively simulating scenarios where not all data attributes are available for analysis. To accomplish this, the selected features are masked with a placeholder value. This masking serves to obscure the original feature values, creating a modified version of the dataset where only a fraction of the original information is accessible.

To address this discrepancy and maintain the necessary input structure, we use a placeholder value to fill in for the missing features. This approach allows us to

effectively simulate the absence of certain features while preserving the original dimensionality of the data. The placeholder acts as a stand-in for the missing information, ensuring that the modified datapoints can still be processed by the model without triggering errors related to input size mismatches.

The selection of the placeholder value in the "Percentage Features" approach is tailored to accommodate the distinct characteristics of different data types, ensuring that the simulated absence of features is effectively represented across varied datasets. Despite the variability in the choice of placeholder values, a common guiding principle is adhered to: the selected placeholder should be markedly distinct from the range of original data values. This significant deviation ensures that the model recognizes these values as placeholders, effectively simulating missing information while minimizing the risk of confusing the placeholder with genuine data.

Binary Data

For binary data, where the features are typically 0s and 1s, the placeholder might be a value outside this binary range, in this approach we choose -1, clearly distinguishing it from legitimate feature values.

Image Data

For image data, consisting of features represented by 8-bit integer values within the range of 0 to 255, selecting an appropriate placeholder for missing data necessitates a choice that stands out distinctly from this original value range. Given the constraints and the nature of the data, we opt to utilize 0 as the placeholder value. This decision is informed by the consideration that, while 0 is indeed within the standard range of pixel intensities, its interpretation as 'no intensity' or 'absolute black' can effectively signify the absence or inapplicability of data in certain contexts. However, it's critical to acknowledge that 0 is a legitimate value within the dataset's natural domain. Therefore, this choice assumes a specific context or preprocessing strategy where employing 0 as a placeholder does not conflict with the analysis or model training objectives, ensuring that its use as a marker for missing features is clearly defined and understood within the experimental framework.

Text Data

In the process of handling missing or unavailable data within our TF-IDF vectorized text data, we've selected 0 as the placeholder value. This choice is informed by the inherent characteristics of TF-IDF scores, where a score of 0 naturally signifies the absence of a term in a document. Using 0 as a placeholder seamlessly aligns with this logic, indicating either the non-presence of a term or the absence of data. This approach simplifies the data preprocessing phase and ensures the integrity of our model's input, facilitating a clear and straightforward interpretation of the TF-IDF vectors. By standardizing on 0 to represent missing information, we maintain a coherent and consistent method for indicating missing or unavailable features, streamlining both the analysis and interpretation processes without the need for negative values.

5.4 Approach 3: Feature Selection

In the "Feature Selection" approach, the implementation strategy diverges from the "Percentage Feature" approach by adopting a more targeted method for feature reduction. Rather than randomly selecting a specific percentage of features to mask with a placeholder, this approach employs mutual information to discern the features with the highest relevance to the target variable. Based on mutual information scores, a predetermined percentage of features exhibiting the highest mutual information are identified and retained for analysis. The remaining features, deemed less informative, are masked with a placeholder value. For the placeholder value, we are using the same we used in the percentage features approach.

5.5 Approach 4: Enumerate Features

In the "Enumerate Features" approach, our primary goal is to explore the full spectrum of potential impacts that missing features might have on a model's ability to accurately identify data membership. This methodical exploration involves initially designating a specified percentage of features within the dataset as missing. This step is crucial as it sets the stage for a comprehensive investigation into how the absence of different subsets of features influences model predictions.

Once these features are identified as missing, the approach systematically generates a variety of scenarios, each corresponding to a different combination of missing features. For every such combination, a new datapoint is created. This process effectively simulates an exhaustive range of possible data conditions, allowing for an in-depth analysis of the model's behavior in response to varying degrees of information absence. At last we use all the data points that are generated and validate if it is in side target model training dataset.

This approach, while offering profound insights into the impact of missing features on model behaviour and vulnerability to Membership Inference Attacks, demands considerable computational resources. This exhaustive method requires generating a new data point for every possible combination of missing features, significantly increasing the computational load as the number of features grows.

Given these computational constraints, applying this approach is most feasible on binary data. Binary data, with features that assume only two states (0 and 1)[image data 0-255/ vectorized text data unlimited), presents a relatively simplified scenario for enumeration. The binary nature of the data significantly reduces the complexity and number of possible combinations of missing features compared to datasets with multi-valued or continuous features. For each feature designated as "missing," there are only two options to enumerate, making the exploration of all possible scenarios computationally more manageable.

This focus on binary data allows for a detailed exploration of model sensitivity to feature omission within a realistic computational framework. By systematically varying the presence of features and observing the model's response, valuable insights can be gained into which features are critical for the model's decision-

making process and, consequently, which might reveal information about data membership. Despite the method's limitation to binary data due to computational demands, the insights derived from this approach contribute significantly to understanding and enhancing data privacy and security in machine learning applications.

Chapter 6

Experiments

In this chapter, we detail the experimental procedures for each approach across various data types, acknowledging that the nature of these data types necessitates distinct procedural adjustments. Despite these data-specific variations, the experimental framework remains consistent across different machine learning models. This consistency ensures that any observed effects on the efficacy of Partial Membership Inference Attacks are attributable to the data type and the applied approach rather than differences in model architecture or training methodology.

During the experiments on partial membership inference attack, each of the described approaches was integrated within the framework of a membership inference attack by [9], with the methodologies specifically applied during the validation phase on the validating data. This means that the techniques—ranging from simulating missing features to selectively adding noise or employing feature selection based on mutual information—were not used in the initial training of the models. Instead, they were applied to the data used for validating the efficacy of the membership inference attacks.

To evaluate the success and impact of Membership Inference Attack, we employ three key metrics: accuracy, precision, and recall. These metrics are chosen for their relevance in determining the effectiveness of Membership Inference Attack, which fundamentally seek to identify whether specific data points were part of a model’s training dataset.

- **Accuracy** measures the overall correctness of the attack, indicating the proportion of true results (both true positives and true negatives) among the total number of cases examined.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision** assesses the accuracy of the positive identifications, indicating the proportion of true positive results in all positive predictions made by the attack.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall** measures the ability of the attack to identify all relevant instances, indicating the proportion of true positive results out of all actual positives in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN}$$

For the purpose of these measurements:

- True Positives (TP) are data points correctly identified by the attack as belonging to the training dataset.
- False Positives (FP) are data points incorrectly identified as belonging to the training dataset but actually are in testing dataset.
- True Negative (TN) are data points correctly identified by the attack as belonging to the testing dataset
- False Negative (FN) are data points incorrectly identified as belonging to the testing dataset but are in training dataset

To accurately measure these metrics, we use data from the target model’s training set to calculate true positives, thereby gauging accuracy, precision, and recall. For measuring false positives, which influence the calculation of precision and recall, we utilize the testing data. This methodology allows for a comprehensive evaluation of the Membership Inference Attack performance, providing insight into both its ability to correctly identify training data points (precision and recall) and its overall accuracy.

This approach ensures that our assessment of Membership Inference Attack is both thorough and nuanced, reflecting the complex dynamics at play in determining whether a data point was part of a model’s training dataset. By adopting this structured evaluation framework, we aim to deepen our understanding of Membership Inference Attack effectiveness across different data types and under various experimental conditions, contributing valuable knowledge to the field of data privacy and security.

6.1 Approach 1: Partial Noise

In the "Partial Noise" approach, two adjustable parameters are pivotal: the level of noise applied and the percentage of features selected for noise addition.

6.1.1 Binary data

In our "Partial Noise" approach tailored to binary data, the experimental setup is designed to explore the effects of varying noise levels on the success of membership inference attacks. Specifically, we examine the impact of different probabilities of bit flipping within the selected features, categorized into mild (0.1), intermediate (0.5), and severe (1.0) noise levels. The experiment commences at the mild noise level, with an initial focus on altering 5% of the features, a decision aimed at introducing a subtle degree of perturbation to the dataset. This approach enables us to assess the initial impact on the attack’s efficacy while preserving the overall structure of the data.

The experimental procedure is characterized by a systematic analysis across these noise levels, each time starting with targeting a small percentage of features and observing the influence on the accuracy of the membership inference attacks. The key objective is to identify the point at which the accuracy of the attack aligns with that of random guessing, indicative of a 50% accuracy rate, suggesting the noise has effectively masked the data to thwart the attack’s success.

For each noise level—mild, intermediate, and severe—the process begins anew, with 5% of the features being subjected to bit flipping, and incrementally increases the proportion of features affected. This methodical escalation continues until the attack’s precision is reduced to chance levels. Through this iterative approach, spanning mild to severe noise applications, we aim to delineate how varying degrees of noise influence the vulnerability of binary data to membership inference attacks, providing insights into the noise thresholds that safeguard data privacy against such analytical breaches.

6.1.2 Image data and Text data

In our investigation of the "Partial Noise" approach applied to both image data and vectorized text data, we adapt the methodology used for binary data by replacing bit flipping with the introduction of Gaussian noise. This modification aims to simulate the type of variability and disturbances that could naturally occur in the processing or acquisition of image and textual data. The experimental framework is defined by two key parameters: the Gaussian noise level and the proportion of the datapoint’s features that the noise affects. Starting with a mild noise level of 0.1, we aim for minimal disruption, applying noise to 5% of the features to introduce a controlled level of uncertainty into the dataset. This initial noise application might be random or follow a specific pattern, depending on the goals of the experiment.

After the initial noise application, we methodically increase the noise-affected feature percentage, closely observing how each increment impacts the effectiveness of Membership Inference Attacks. This gradual increase proceeds until reaching a point where the Membership Inference Attack accuracy is notably reduced, approaching a 50% success rate, which would indicate a significant decline in the attack’s effectiveness to the level of random guessing. This threshold marks the noise level at which data obfuscation becomes effective in protecting against Membership Inference Attack.

If further investigation into higher noise levels is warranted, the noise level is increased beyond 0.1, and the iterative process of incrementally increasing the noise-affected feature percentage begins anew. Each iteration seeks to explore a different intensity of noise application, starting again from a low percentage of noised features and gradually increasing it, to meticulously evaluate its influence on the accuracy of Membership Inference Attack.

This detailed experimental approach facilitates a comprehensive analysis of how different levels of Gaussian noise, applied to varying proportions of features, affect the susceptibility of image and text data to Membership Inference Attack. By systematically exploring these dimensions, the aim is to uncover the optimal

balance between ensuring data privacy and maintaining the practical utility of the data for machine learning applications.

6.2 Approach 2: Percentage Features

In the "Percentage Features" approach all types of data have the same, we initially randomly select 95% of the features, to validate the Membership Inference Attack. We then decrease the percentage of features used for validation by 5% in each subsequent iteration, continuing this process until the attack becomes random guessing.

6.3 Approach 3: Feature Selection

In the "Feature Selection" approach, we start by employing mutual information to identify the top 95% of features based on their relevance. With each iteration, we reduce this subset by 5%, focusing on a progressively smaller set of top features, and continue this process till it has the same amount of features that cause percentage Features Approach to start random guessing and compare the result of both approaches.

6.4 Approach 4: Enumerate Features

In our "Enumerate Features" approach, due to computational constraints, the focus is exclusively on binary data, where each feature can exist in one of two states. This limitation arises from the exponential increase in computational complexity when enumerating possible feature combinations, even with a small fraction of features treated as missing. Specifically, we limit our exploration to just 0.01 percent of features regarded as missing, a choice dictated by the necessity to manage computational demands.

For instance, in a dataset where each data point has 600 features, treating 1% as missing equates to having 6 features absent. Given that each binary feature can assume two states, this scenario results in $64 \cdot 2^6$ different combinations for a single data point. When considering the need to apply this enumeration across numerous data points within the dataset, the computational task becomes exceptionally challenging. This complexity underscores why the "Enumerate Features" approach is applied with such a narrow scope and emphasizes the significant computational resources required to explore feature enumeration even on a small scale within binary datasets.

Chapter 7

Results

In this section we examine the results of applying different methods to different machine learning models, focusing on precision, accuracy, and recall analysis as our key performance metrics to ensure fair and ongoing comparisons ever Let us hold it with caution. This standardized approach allows us to examine the effects of each method directly, eliminating differences caused by changes in the size or volume of the data set. In conducting the attack, we ensure uniformity across all models involved—the target, shadow, and attack models—all utilize the same machine learning architecture. Moreover, for each attack conducted across various machine learning models and data types, we standardly employ two shadow models as part of the attack strategy.

7.1 Logistic Regression

As mentioned in the methodology section, we’ve selected logistic regression as the starting point for our exploration into Partial Membership Inference Attacks, primarily due to its straightforward nature which facilitates a clear foundational understanding of how these attacks function. The simplicity and transparency of logistic regression make it an excellent model for initial investigations into Membership Inference Attack, allowing for easier analysis and interpretation. However, it’s critical to acknowledge that the inherent characteristics of logistic regression—namely its lower complexity and its natural resistance to overfitting—may limit its ability to fully demonstrate the vulnerabilities that might be exploited by Membership Inference Attack in more sophisticated models. Overfitting is a crucial element that influences the effectiveness of Membership Inference Attack, as models that closely fit their training data tend to be more revealing about that data. Since logistic regression is designed to avoid overfitting, it may not be as prone to Membership Inference Attack as models with higher complexity. Therefore, while logistic regression provides a solid foundation for beginning our study, expanding our research to encompass more complex models is imperative for acquiring a deeper and more comprehensive insight into the vulnerabilities posed by Membership Inference Attack across a broader array of machine learning architectures.

7.1.1 Approach 1: Partial Noise

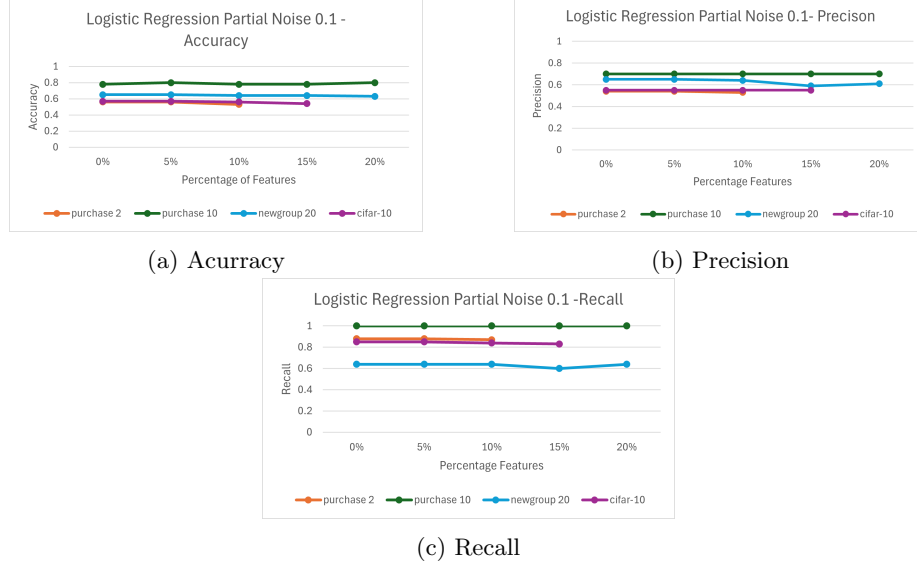


Figure 7.1: Performance metrics for mild Partial Noise approach on Logistic Regression

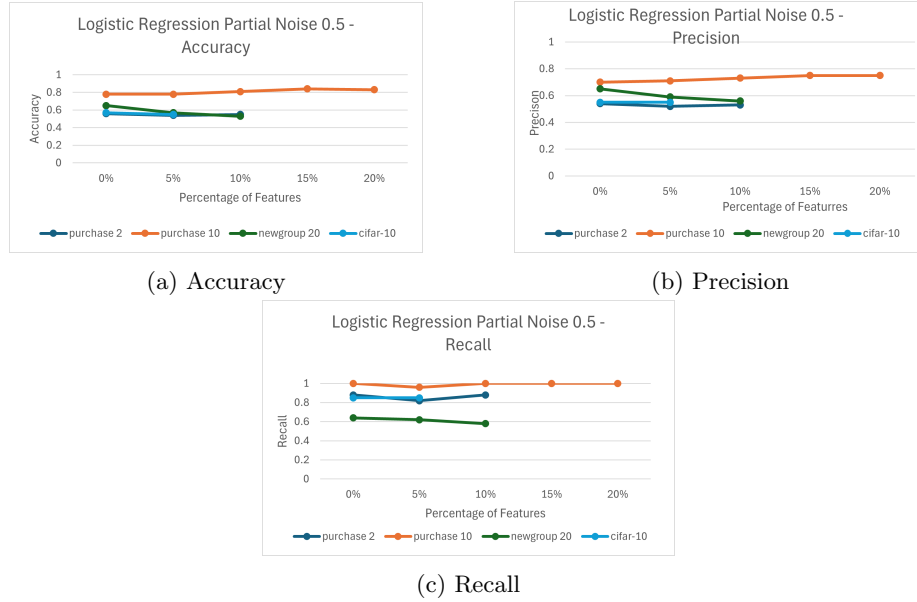


Figure 7.2: Performance metrics for intermediate Partial Noise approach on Logistic Regression

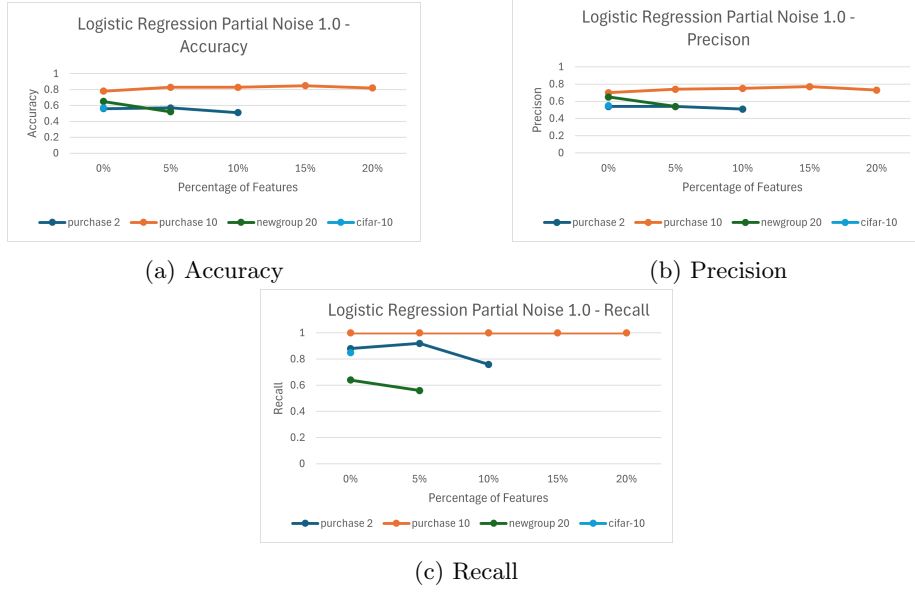


Figure 7.3: Performance metrics for severe Partial Noise approach on Logistic Regression

In our exploration of the Partial Noise approach applied to logistic regression, we delve into the impact of introducing noise at various intensities—mild (0.1 noise level), intermediate (0.5 noise level), and severe (1.0 noise level)—on different portions of features within a data point before evaluating its membership. This comprehensive analysis aims to understand how noise addition affects the key metrics of Membership Inference Attacks—accuracy, precision, and recall—across diverse data types, namely binary, image, and text data.

Figure 7.1 demonstrates that adding a mild level of noise to a subset of features does not significantly alter the attack’s effectiveness across any of the data types examined. This observation suggests that a low degree of noise is insufficient to mask the features to a point where the model’s ability to infer membership is compromised. However, the scenario shifts as we increase the noise level to intermediate and severe, as depicted in Figures 7.2 and 7.3. At these higher noise levels, even when noise is applied to merely 5% of the features, there’s a noticeable decline in the attack’s performance on text and image data. This significant drop highlights the sensitivity of these data types to noise, suggesting that the added noise disrupts the model’s predictive cues used for membership inference.

Contrastingly, the attack’s efficacy on binary data remains unaffected by the addition of noise, regardless of its intensity. This unchanged performance, even in the face of intermediate and severe noise levels, may reflect the unique properties of binary data or the characteristics of models trained on such data.

7.1.2 Approach 2: Percentage Features

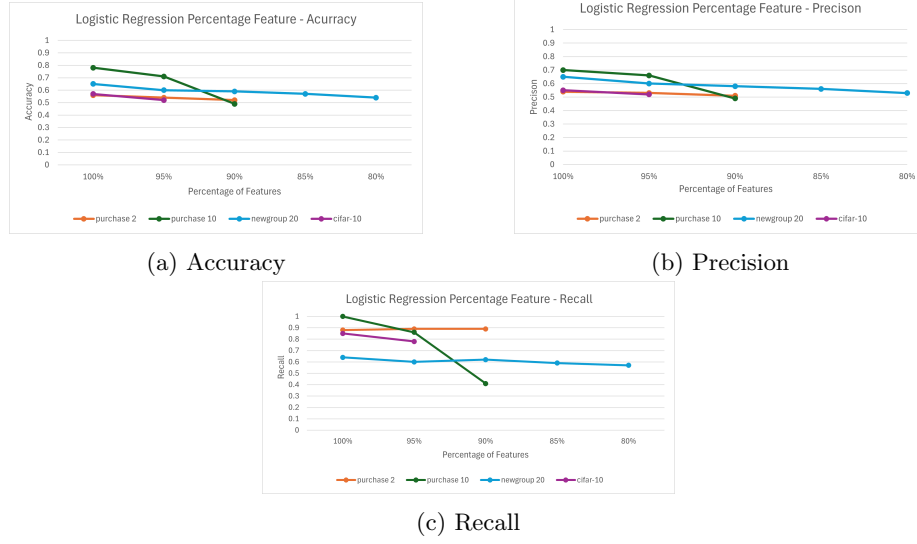


Figure 7.4: Performance Metrics for Percentage features approach on Logistic Regression

In our examination of the Percentage Features approach applied to logistic regression, the results shed light on how the attack's effectiveness is impacted by the omission of parts of a data point. Specifically, Figure 7.4 reveals a notable divergence in attack performance across different data types when only 90% of the features are present.

For binary and image data, the attack's accuracy descends to the level of random guessing once 10% of the features are missing. This indicates a significant reduction in the attack's effectiveness, suggesting that the absence of a relatively small portion of features severely compromises the ability to infer membership in the training dataset for these types of data. Conversely, the attack's performance on text data exhibits remarkable resilience under the same conditions, remaining stable even with 20% of features missing.

7.1.3 Approach 3: Feature Selection

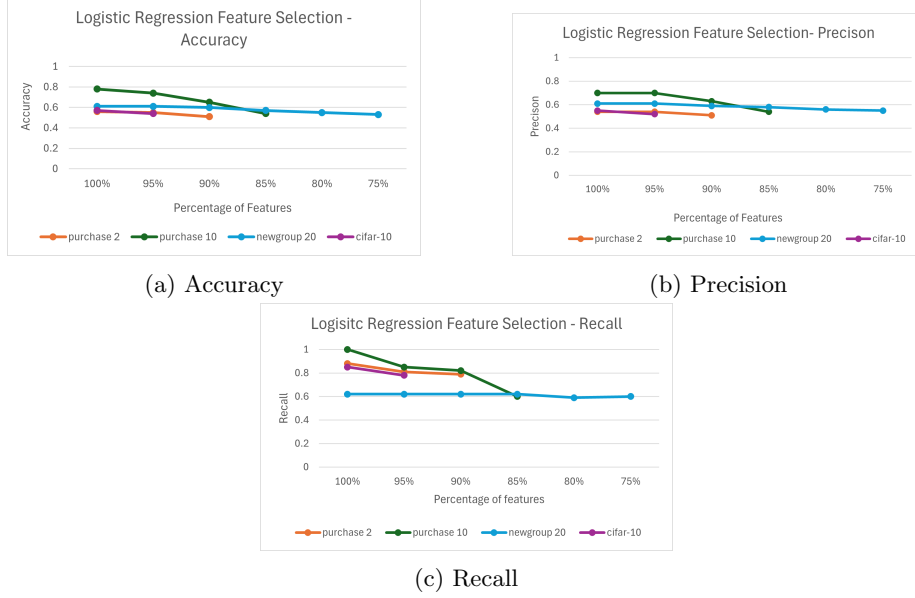


Figure 7.5: Performance Metrics for Feature Selection approach on Logistic Regression

In our experiment utilizing the feature selection approach applied to logistic regression, the objective was to surpass the performance observed with the percentage features approach. According to Figure 7.5, the performance of the attack remains largely stable across most datasets, with a notable exception observed in the "purchase 10" dataset. Contrary to the percentage features approach, where the attack's effectiveness dwindles to random guessing with 90% of features available, the feature selection approach demonstrates a slight improvement. Specifically, it's only when the availability of data drops to 85% that we observe the performance of the attack declining to the level of random guessing. This distinction underscores the potential for the feature selection approach to maintain higher attack efficacy under conditions of reduced feature availability, suggesting a more robust resilience against feature omission compared to the baseline percentage features strategy.

7.1.4 Approach 4: Enumerate Features

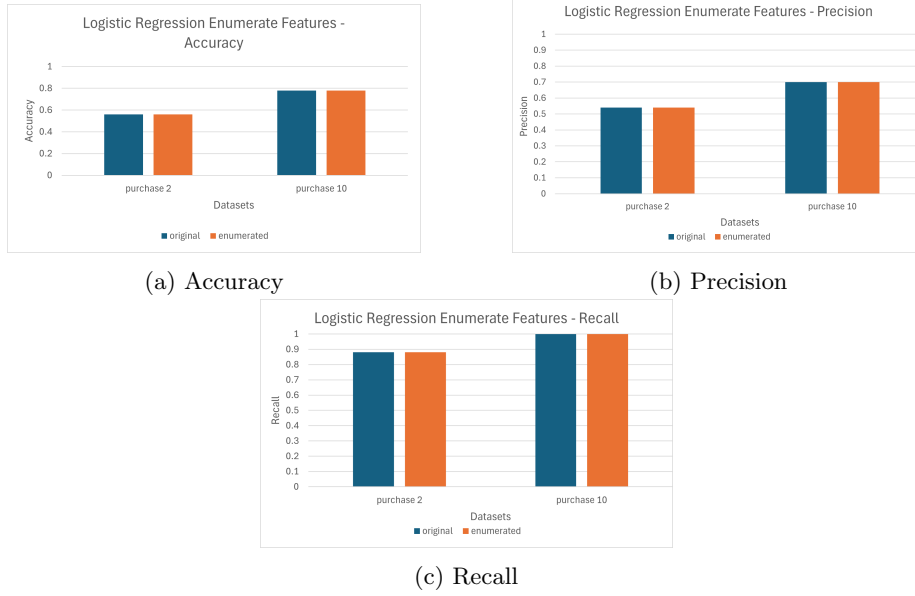


Figure 7.6: Performance Metrics for Feature Selection approach on Logistic Regression

In our experiment with the enumerate features approach applied to logistic regression, our goal was to exhaustively explore every potential configuration of missing features. However, due to the significant computational demands of this method, we were constrained to examining scenarios with only 1% of features missing, systematically enumerating this small fraction. When comparing the outcomes of this constrained enumeration to the baseline attack performance, as depicted in Figure 7.6, we observed no discernible degradation in the effectiveness of the attack. This result aligns with expectations, as the omission of merely 1% of the features is unlikely to substantially impact the model’s ability to predict membership, suggesting that such a minimal level of feature absence does not meaningfully alter the attack’s dynamics or reduce its success rate.

7.2 MultiLayer perception neural network

As outlined in the methodology section, the Multilayer Perceptron (MLP) is specifically chosen for Partial Membership Inference Attacks due to its inherent complexity. This complexity, while beneficial for capturing intricate patterns in data, predisposes MLPs to overfitting unless careful precautions are taken. Overfitting in the context of MLPs can significantly enhance the success rate of Membership Inference Attacks, as models that overfit their training data tend to reveal more information about it, making them more vulnerable to such attacks. Thus, the selection of MLPs serves as a strategic choice to investigate the nuances of attack success in scenarios where the model’s complexity could

inherently compromise data privacy.

7.2.1 Approach 1: Partial Noise

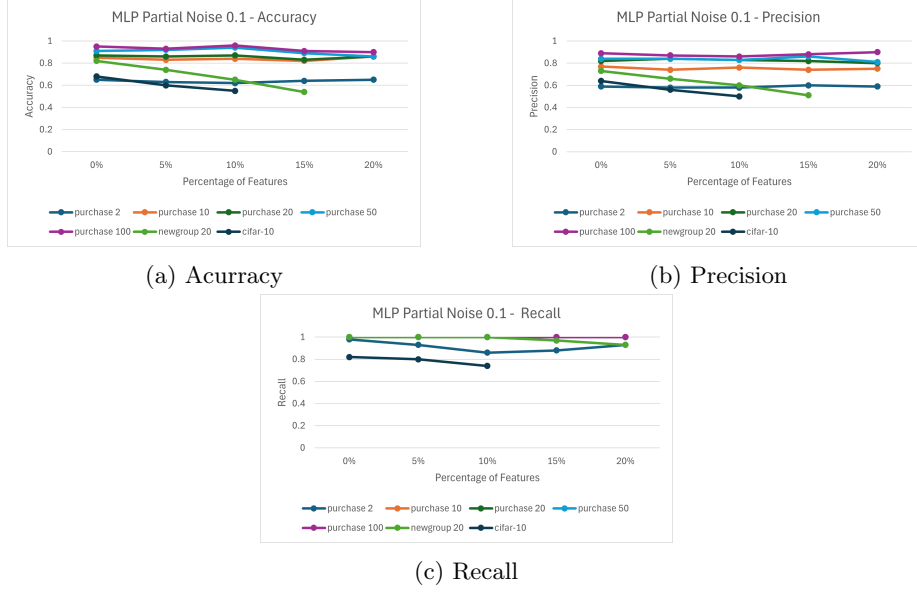


Figure 7.7: Performance metrics for mild Partial Noise approach on MLP

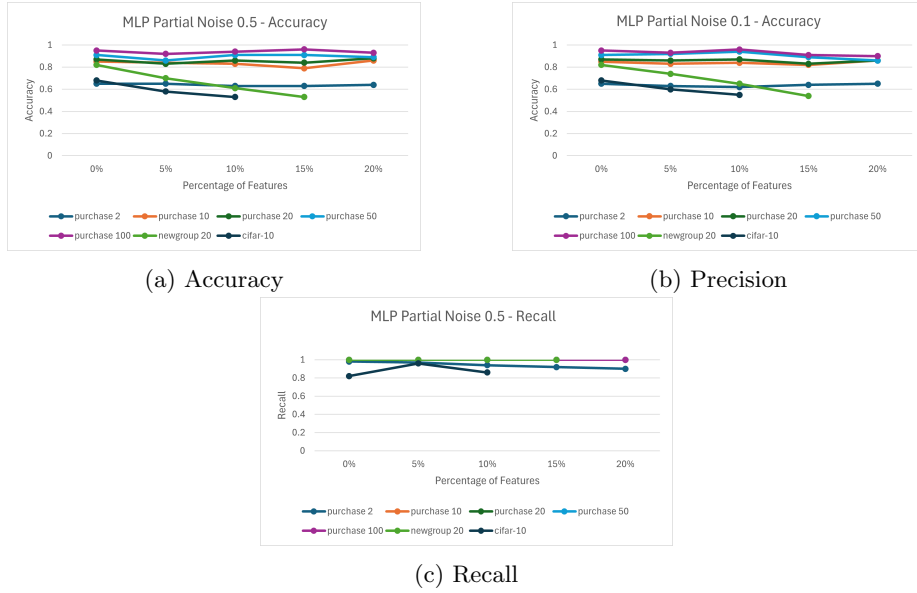


Figure 7.8: Performance metrics for intermediate Partial Noise approach on MLP

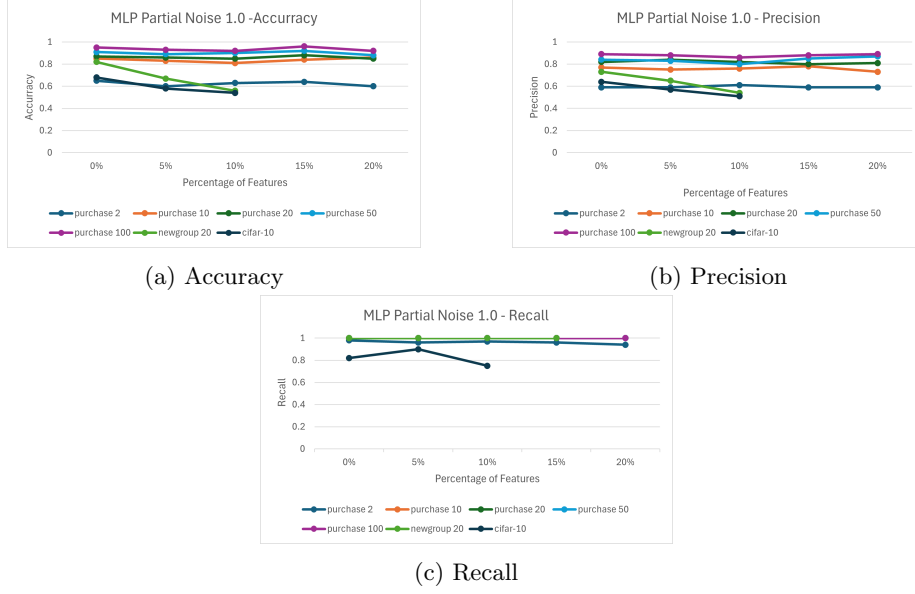


Figure 7.9: Performance metrics for severe Partial Noise approach on MLP

In our experiments employing the partial noise approach on Multilayer Perceptrons (MLP), we observed a pattern consistent with the results from applying it to logistic regression. Specifically, the performance of Membership Inference Attacks on binary data remained unaffected by the introduction of noise, irrespective of whether the noise was at intermediate or severe levels. This resilience suggests that binary data, when processed by MLPs, is less sensitive to noise perturbations, possibly due to the inherent characteristics of binary datasets or the robustness of MLP architectures in handling such data. Conversely, for text and image data, the attack's effectiveness experienced a decline even when only mild levels of noise were introduced, at severe level of noise both text and image data start to random guess while we only introduced noise to 10% of features.

7.2.2 Approach 2: Percentage Features

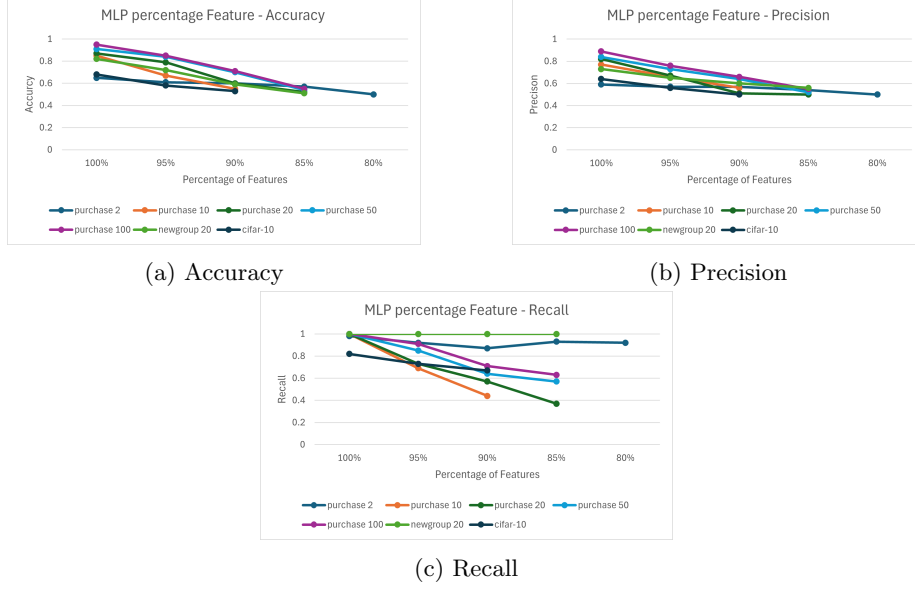


Figure 7.10: Performance Metrics for Percentage features approach on MLP

In our exploration of the Percentage Feature approach applied to Multilayer Perceptrons (MLP), we identified a distinctive pattern in the resistance of binary data to Membership Inference Attacks as the number of classes increases. Specifically, in scenarios involving binary data with 10 classes, the attack’s efficacy dwindles to that of random guessing when as little as 10% of features are missing. Interestingly, for binary datasets encompassing more than 10 classes, the threshold at which the attack’s accuracy drops to random guessing rises to 15% missing features. This indicates a heightened resilience against feature omission in binary data with a larger class count when processed by MLPs.

Contrasting these findings with those from logistic regression experiments, we observed that the performance of the attack on text data does indeed diminish under the MLP model, diverging from the stability noted in logistic regression scenarios. The attack’s effectiveness against text data reduces to random guessing with 15% of the features missing. Meanwhile, image data demonstrates a pronounced vulnerability, with the attack performance hitting the random guessing benchmark as soon as 5% of features are absent.

7.2.3 Approach 3: Feature Selection

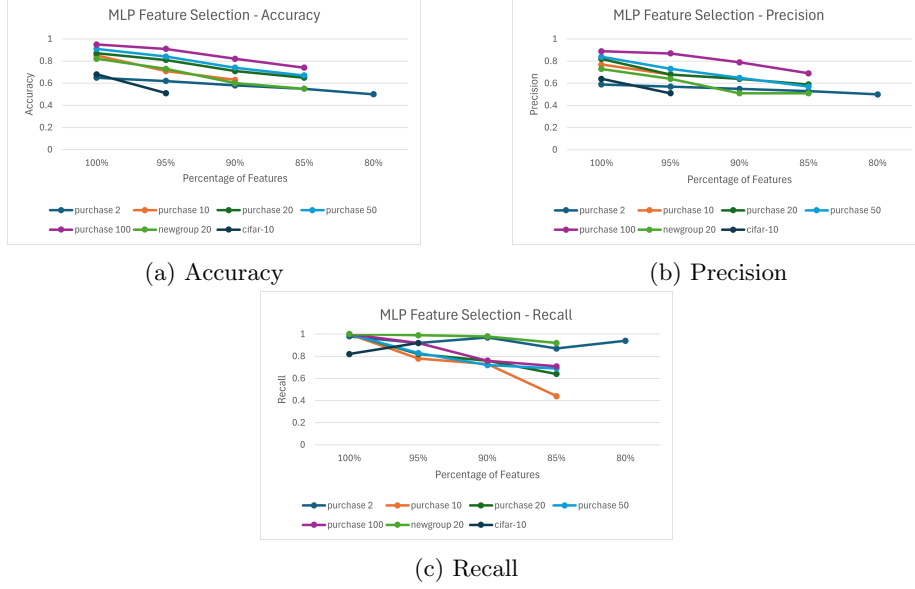


Figure 7.11: Performance Metrics for Feature Selection approach on MLP

In our experimentation with the feature selection approach applied to Multilayer Perceptrons (MLP), a noticeable improvement in attack performance against binary and text data was observed compared to the percentage feature approach. Specifically, when examining scenarios with 15% of features missing, the feature selection approach did not deteriorate to random guessing, highlighting its superior resilience. This contrasts with the percentage feature approach, where the attack’s effectiveness had already diminished to random guessing under the same conditions for these data types. On the other hand, the performance against image data remained consistent between the two approaches, with both starting to approximate random guessing when 5% of the features were missing. This indicates that for binary and text data, strategically selecting which features to omit—based on their informational value—can significantly enhance the robustness of MLPs against Membership Inference Attacks, whereas image data’s susceptibility to such attacks appears relatively unaffected by the feature selection strategy.

7.2.4 Approach 4: Enumerate Features

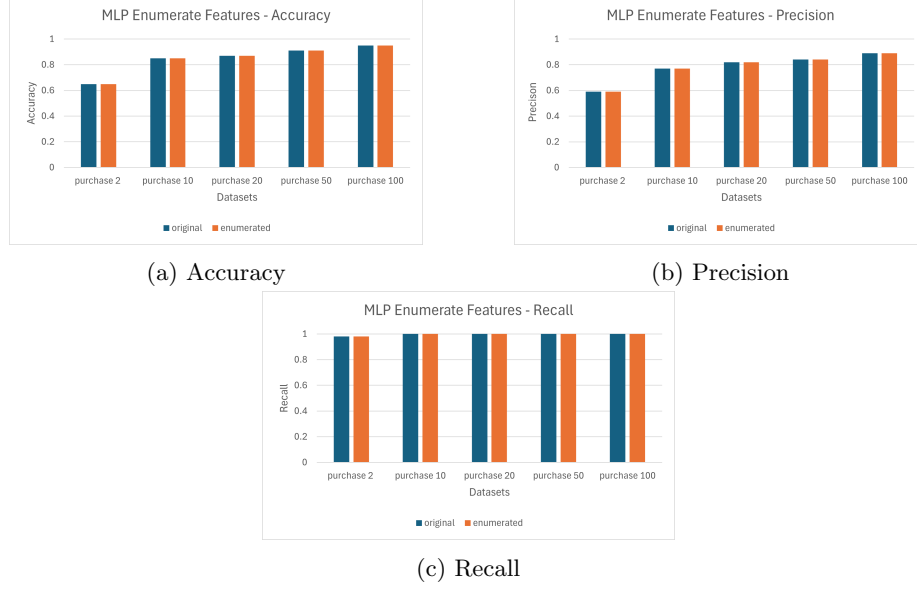


Figure 7.12: Performance Metrics for Feature Selection approach on MLP

In our experiment with the enumerate features approach applied to MLP, our goal was to exhaustively explore every potential configuration of missing features. However, due to the significant computational demands of this method, we were constrained to examining scenarios with only 1% of features missing, systematically enumerating this small fraction. When comparing the outcomes of this constrained enumeration to the baseline attack performance, as depicted in Figure 7.12, we observed no discernible degradation in the effectiveness of the attack. This result aligns with expectations, as the omission of merely 1% of the features is unlikely to substantially impact the model's ability to predict membership, suggesting that such a minimal level of feature absence does not meaningfully alter the attack's dynamics or reduce its success rate.

Chapter 8

Discussion

8.1 Limitations

Several limitations have arisen in this work, especially in relation to the choice of materials and the percentage of materials used. A major limitation stems from our approach to modeling missing items through placeholders, rather than actually reducing the number of items used in experiments. This approach was necessary because of the lack of data from the models producing a handling with fewer resources than originally trained. Furthermore, identifying locations that fall exactly where machine learning models search severely limits our ability to reduce the number of features used in our search. Furthermore, the use of placeholders introduces an element of inconsistency in the results of membership theory attacks, where attacks on preconceived states sometimes succeed and times another fails. This variation presents a challenge to draw conclusive conclusions from the results.

8.2 Future Work

In the experimental section, we discussed the considerable computational complexity associated with the enumerate approach, highlighting the need for strategies to lessen this computational burden while maintaining the efficacy of the approach. Additionally, the nature of Membership Inference Attacks (MIA), which often rests on a set of assumptions that may not always hold true in real-world scenarios, presents another challenge. These assumptions, such as the attacker's need to access the full data point to determine its presence in the training dataset of a machine learning model, limit the applicability of Membership Inference Attack in practical situations. To address these limitations, an integration of our methodology with insights drawn from related work could offer a promising path forward. By adopting and adapting methodologies from studies such as those by [3] [4] [2] [7] we can work towards a more generalized framework for Membership Inference Attack. These referenced works introduce innovative approaches to understanding and mitigating the risks associated with Membership Inference Attack, providing a valuable foundation upon which to

build.

The goal would be to develop a refined MIA methodology that effectively reduces the reliance on unrealistic assumptions, such as the necessity for complete data points for attack execution. By incorporating strategies from the related literature, there’s potential to devise approaches that are not only computationally more efficient but also more relevant to real-world settings. This could significantly enhance the feasibility of Membership Inference Attack as a practical attack vector, broadening their applicability and relevance in the assessment of privacy vulnerabilities in machine learning models.

8.3 Conclusion

In this research, we aimed to explore the boundaries of Membership Inference Attacks (MIA), specifically investigating whether such attacks could successfully determine the inclusion of a data point in a target model’s training dataset when only partial information from that data point is available. Our experiments spanned across different data modalities, including text, binary, and image datasets, to provide a comprehensive understanding of MIA’s applicability and effectiveness across diverse data types.

We embarked on a series of experimental approaches to probe the nuances of MIAs under these constrained scenarios. This included the Partial Noise Approach, where selected portions of data were subjected to noise, and the Percentage Feature Approach, which examined the impact of missing features on the efficacy of MIAs. To refine and potentially enhance the attack strategies, we utilized mutual information to select the most informative features for our analysis. This methodological adjustment showed promise, yielding some improvement in attack performance compared to the baseline Percentage Feature Approach, particularly highlighting the benefits of strategic feature selection.

Moreover, we investigated an exhaustive strategy through the Enumerate Features Approach, aiming to account for every possible configuration of missing features. While theoretically optimal in preserving attack performance, this approach faced significant computational barriers that precluded a full-scale evaluation.

Throughout these explorations, testing was conducted across text, binary, and image data, ensuring a broad assessment of the attack’s versatility and identifying specific challenges and opportunities unique to each data type.

In conclusion, our research offers valuable insights into the capabilities and limitations of Membership Inference Attacks when faced with incomplete data across various data forms. The findings underscore the potential for methodological enhancements in MIAs through informed feature selection and highlight critical computational challenges in exhaustive enumeration strategies. This work advances the discussion on maintaining data privacy in machine learning models and emphasizes the need for ongoing efforts to develop scalable, effective techniques for privacy preservation in diverse data landscapes.

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Appendix

Gitlab link: <https://git.cs.bham.ac.uk/projects-2023-24/cx1234>.